Instantaneous Magnetic Field Distribution in Permanent Magnet Brushless dc Motors, Part IV: Magnetic Field on Load

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***Abstract*—An analytical technique is developed for predicting the instantaneous magnetic field distribution in the airgap-permanent magnet region of radial-field topologies of brushless dc motors equipped with a surface mounted magnet rotor and operating under any specified load condition. It accounts implicitly for the stator winding current waveform and the effect of stator slot openings. The 2-dimensional field analysis, in polar coordinates, combines the armature reaction field component, derived from an analytical prediction of the current waveform, with the open-circuit field component produced by the magnets, the resultant field being predicted under any commutation strategy through the analytical determination of the relative temporal and spatial position of the two field components. The predicted open-circuit, armature reaction, and load field distributions all show excellent agreement with results from corresponding finite element analyses.**

I. Introduction

One consequence of the adoption of electronically commutated brushless dc motors in motion control systems is the increased possibility of resonances between the electromagnetic exciting forces and the stator structure as well as an increase in the emitted acoustic noise due to the fact that the phase current waveforms contain significant harmonics [1]. To predict the vibrational and acoustic behavior it is necessary to analyze the airgap field distribution under any specified load condition.

Although several papers have been published on the calculation of the airgap magnetic field of brushless dc motors, e.g., [2], [3], none have been concerned specifically with the analysis of the magnetic field on load. However, in parts I–III [5]–[7] of this series of papers analytical solutions for the open-circuit and armature reaction field components have been derived for radial-field motor topologies in which the magnets are mounted adjacent to the airgap. These analyses are combined in accordance with the time variation of the relative permeance and the back-emf and current waveforms to enable the resultant instantaneous magnetic field to be deduced under any load condition and commutation strategy.

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II. Spatial and Temporal Relationship Between Open-Circuit and Armature Reaction Field and the Relative Permeance

Fig. 1 indicates by means of a flow-chart those parameters and performance factors for which the authors have developed analytical techniques for their evaluation. Central to all of them is the analysis of the magnetic field in the airgap region. However, by assuming infinite permeability for the stator and rotor iron, the analysis is reduced to a linear problem, and the instantaneous field distribution under any specified load condition can be obtained by superposition of the open-circuit and armature reaction field components, i.e.,

$$B_{load}(\alpha, r, t) = B_{open\text{-}circuit}(\alpha, r, t) + B_{armature\text{-}reaction}(\alpha, r, t) \quad (1)$$

where

$$B_{open\text{-}circuit}(\alpha, r, t) = B_{magnet}(\alpha, r, t)\tilde{\lambda}(\alpha, r) \quad (2)$$

and

$$B_{armature\text{-}reaction}(\alpha, r, t) = B_{winding}(\alpha, r, t)\tilde{\lambda}(\alpha, r). \quad (3)$$

As mentioned in part III [7], the open-circuit field distribution is calculated from the product of the field produced by the magnets when the stator slot openings are neglected and the relative permeance function of the slotted airgap region. In a similar manner the armature reaction field distribution is calculated from the product of the magnetic field produced by the stator windings when stator slotting is neglected and the relative permeance function. The calculation of the relative permeance function and the field components produced by the magnets and the windings, i.e., $\tilde{\lambda}(\alpha, r)$, $B_{magnet}(\alpha, r, t)$, and $B_{winding}(\alpha, r, t)$, are described fully in parts I–III [5]–[7].

To obtain the instantaneous magnetic field distribution under any load condition and commutation strategy it is necessary to pre-determine the relative position of the open-circuit and armature reaction field components and the permeance function according to the instantaneous rotor position and the corresponding phase currents as discussed below.

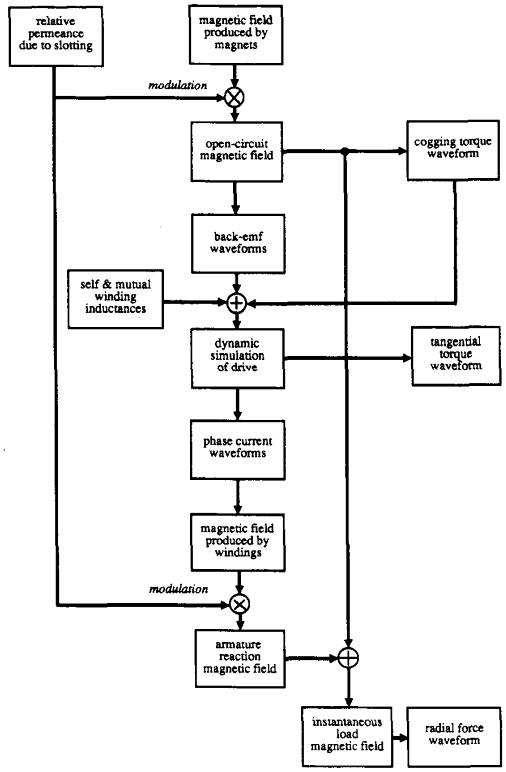

Fig. 1. Flow-chart for analytical electromagnetic and related analyses.

A. Reference Direction of Rotation of Rotor

It is assumed that the direction of rotation of the rotor is anti-clockwise, as shown in Fig. 2.

B. Reference Polarity of Phase Currents and Winding Phase Sequence

The reference direction of the current flowing in phase A of Fig. 2 is such that:

1) In the conductors of coil side A current flows into the plane of the paper, whilst in the conductors of coil side A′ it flows out of the plane of the paper.
2) The polarity of the mmf produced by the windings is determined by Fleming's right-hand rule.

The phase sequence is A, B, C, such that as the rotor rotates the emf induced in phases B and C is, respectively, 120° elec. retarded and 120° elec. advanced in time with respect to the emf induced in phase A. This definition ensures that the time phase sequence is the same as the spatial phase sequence.

The reference position for the axis of phase A is such that the initial spatial angular position $\alpha = 0$ is defined at the axis of phase A, whilst the current waveform of phase A is such that following the commutation event at time $t = 0$ the current i_a has a positive polarity, as shown in Fig. 3.

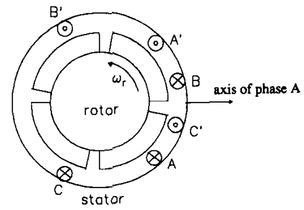

Fig. 2. Sequence of phases A, B, and C in space.

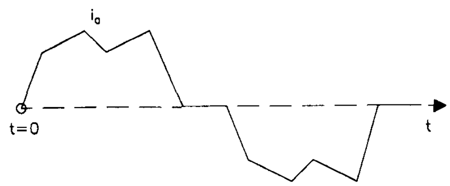

Fig. 3. Definition of initial time for current waveform of phase A.

C. Reference Position of Field Components Produced by Magnets and Windings and the Permeance Function

In part I [5], the solution for the magnetic field produced by the permanent magnets was with reference to $\theta = 0$, i.e., the axis of a magnet pole, whose direction of magnetization is along the positive radial direction of the polar coordinate system, i.e.,

$$B_{magnet}(\theta, r) = \sum_n B_n \cos np\theta. \tag{4}$$

Similarly, the solution obtained in part II [6] for the magnetic field produced by the windings was with reference to $\alpha = 0$, which corresponds to the axis of the phase A winding, i.e.,

$$B_{winding}(\alpha, r, t) = \sum_u \sum_v B_{uv} \sin(up\omega_r t - v_u\alpha + \theta_u). \tag{5}$$

However, the reference position $\alpha = 0$ for the relative permeance function, obtained in Part III, was with respect to the center of a stator slot opening, i.e.,

$$\tilde{\lambda}(\alpha, r) = \sum_\mu \tilde{\Lambda}_\mu \cos \mu Q_s\alpha \tag{6}$$

D. Relationship Between Time and Rotor Angular Position

The relationship between time and the rotor angular position can be determined from a dynamic simulation of a brushless drive system, as described in [4]. However, if the speed of the rotor can be assumed to be constant, then for steady-state operation the relationship is obtained simply from the preceding definitions, viz.

$$\alpha = \omega_r t = \frac{2\pi n_r}{60} t \quad \text{(mechanical rad.)} \tag{7}$$

or

$$\alpha = \omega_r t = \frac{360 n_r}{60} t = 6n_r t \quad \text{(mechanical degree)} \tag{8}$$

which yields:

$$t = \frac{\alpha}{6n_r} \tag{9}$$

where n_r is the speed of the rotor (r/min).

E. Angular Displacement Between the Axis of Phase A Winding and the Permeance Function

The angle between the airgap permeance function and the winding mmf distribution depends on the particular winding arrangement of a motor. As stated in (C), the reference for the permeance function is a stator slot axis. If the reference is redefined to coincide with the axis of phase A stator winding, this may no longer be a stator slot axis. However, in general, there are only two possibilities, viz:

1) When the winding pitch is an odd integer of the slot pitch, as shown in Fig. 4(a), in which case the axis of the coils of phase A coincides with the center of a stator tooth.
2) When the winding pitch is an even integer of the slot pitch, as shown in Fig. 4(b), in which case the axis of phase A coincides with the center of a stator slot.

A 3-phase permanent magnet brushless dc motor generally employs a concentrated stator winding which can have either nonoverlapping or overlapping end-windings. If it is a nonoverlapping winding the winding pitch is equal to one slot pitch, whilst the winding pitch of an overlapping winding is three slot pitches. Therefore in both cases the axis of the coils of phase A will coincide with the center of a stator tooth.

Therefore if the magnetic field produced by the windings is defined by (5) the relative permeance function is given by

$$\tilde{\lambda}(\alpha, r) = \sum_{\mu} \tilde{\Lambda}_\mu \cos \mu Q_s(\alpha + \alpha_{sa}) \tag{10}$$

where

$$\alpha_{sa} = \begin{cases} \textit{half slot pitch} = \dfrac{\pi}{Q_s} & \text{for case 1} \\ 0 & \text{for case 2} \end{cases} \tag{11}$$

i.e.,

$$\tilde{\lambda}(\alpha, r) = \begin{cases} \sum_{\mu} (-1)^\mu \tilde{\Lambda}_\mu \cos \mu Q_s \alpha & \text{for case 1} \\ \sum_{\mu} \tilde{\Lambda}_\mu \cos \mu Q_s \alpha & \text{for case 2} \end{cases} \tag{12}$$

G. Relative Position Between Axis of Phase A Windings and Permanent Magnets

If the open-circuit airgap field produced by the permanent magnets is given by

$$B_{open\text{-}circuit}(\theta, r) = \tilde{\lambda}(\alpha, r) B_{magnet}(\theta, r) = \tilde{\lambda}(\alpha, r) \sum_n B_n \cos np\theta \tag{13}$$

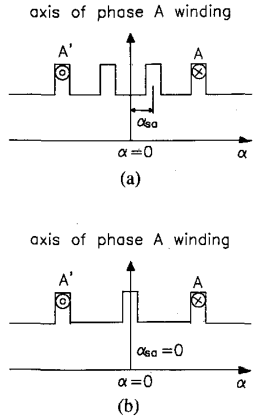

Fig. 4. Relative positions between axes of phase windings and permeance. (a) The winding pitch is an odd integer of the slot pitch. (b) The winding pitch is an even integer of the slot pitch.

where $\theta = \alpha - \alpha_{ma}$, α being the angular position with respect to the axis of the phase A winding, and α_{ma} being the angle between the axes of the magnetic field components due to the phase A mmf and the magnet mmf, then from the Appendix, the induced emf per phase is obtained as:

$$e = \sum_n E_n \sin np\alpha_{ma} \tag{14}$$

where

$$\alpha_{ma} = \omega_r t + \theta_o \tag{15}$$

and θ_o depends on the commutation angle which can be determined as follows.

In a conventional bipolar-fed 3-phase brushless dc motor the commutation of the current in a particular phase is normally delayed by an angle $\pi/6$ (elec. rad.) after the zero crossing of its induced emf, as described in [4] and shown in Fig. 5, positive motoring current being against the emf. Therefore if the phase current is given by

$$i_a = \sum_u I_u \sin u(p\omega_r t + \theta_u) \tag{16}$$

where $i_{a|_{t=0}} = 0$, as indicated in Fig. 5, then for normal commutation

$$e_a = \sum_n E_n \sin n\left(p\omega_r t + \frac{\pi}{6} - \pi\right) \tag{17}$$

whilst for advanced or retarded commutation

$$e_a = \sum_n E_n \sin n\left(p\omega_r t - \alpha_c + \frac{\pi}{6} - \pi\right) \tag{18}$$

where α_c, the commutation angle, >0 for advanced commutation and <0 for retarded commutation.

On comparing (18) with (36) in the Appendix, the following relationship holds:

$$\alpha_{ma} = \omega_r t - \frac{\alpha_c}{p} - \frac{5\pi}{6p}. \tag{19}$$

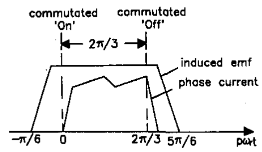

Fig. 5. Relative phase of phase current and back-emf waveforms under normal commutation.

Hence, θ_o is determined from (15) and (19) as:

$$\theta_o = -\frac{\alpha_c}{p} - \frac{5\pi}{6p}. \tag{20}$$

Therefore if the magnetic field produced by the stator windings is given by (5), the magnetic field produced by the magnets is now given by

$$B_{magnet}(\alpha, r, t) = \sum_n B_n \cos np(\alpha - \alpha_{ma}) \tag{21}$$

with α referenced to the axis of the phase A windings.

IV. Instantaneous Field Distributions

In summary the instantaneous field distribution under any specified load condition is calculated from (1)–(3), in which

$$B_{magnet}(\alpha, r, t) = \sum_{n=1,3,5,\cdots,}^{\infty} B_n \cdot \cos np\left(\omega_r t - \alpha - \frac{\alpha_c}{p} - \frac{5\pi}{6p}\right) \tag{22}$$

$$B_{winding}(\alpha, r, t) = \sum_{u=1,2,3,\cdots}^{\infty} \sum_{c=0,\pm 1,\pm 2,\cdots,}^{\infty} B_{uv} \cdot \sin\left[u(p\omega_r t) + v_u\alpha + \theta_u\right] \tag{23}$$

and

$$\tilde{\lambda}(\alpha, r) = \sum_{\mu=0,1,2,\cdots,}^{\infty} \tilde{\Lambda}_\mu \cos \mu Qs(\alpha + \alpha_{sa}) \tag{24}$$

where

$$v_u = \pm p(6c - \{\pm\}u)$$

$$\alpha_{sa} = \begin{cases} \dfrac{\pi}{Qs} & \text{for winding pitch} = \text{odd integer of slot pitch} \\ 0 & \text{for winding pitch} = \text{even integer of slot pitch} \end{cases} \tag{26}$$

and θ_u is the phase angle of the harmonics in the phase current:

$$i_a = \sum_{u=1,2,3,\cdots,}^{\infty} I_u \sin u(p\omega_r t + \theta_u) \tag{27}$$

where $i_{a_t} = 0$, ω_r is the angular speed of rotation of the rotor; and α is the angular position around the airgap, where $\alpha|_{t=0} = 0$ and $\alpha = 0$ correspond with the axis of phase A stator winding. The amplitudes of B_n, B_{uv}, and $\tilde{\Lambda}_\mu$ are calculated by the methods given in parts I, II [5], [6], [7], and III, respectively, so that the instantaneous magnetic field under any load condition and commutation strategy can then be determined.

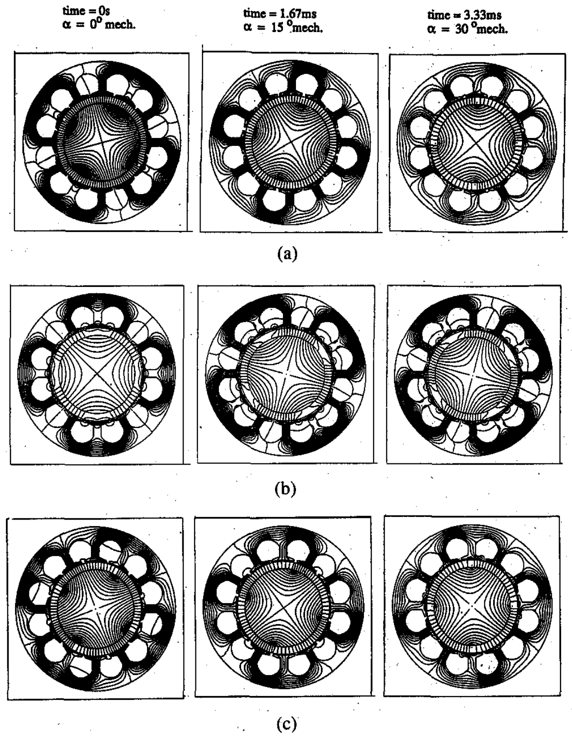

Fig. 6. Typical instantaneous field distributions (1500 r/m). (a) Open circuit field. (b) Armature reaction field. (c) Resultant field.

V. Comparison of Predictions with Finite Element Calculations

Fig. 6 shows instantaneous field distributions derived from finite element analyses of the internal rotor radial-field motor whose parameters are given in Table II of part I [5] and whose mesh discretization is shown in Fig. 7 of part III [7]. The results are for a rotor speed of 1500 rpm, and span a time interval of 3.333 ms, which corresponds to $\frac{1}{6}$ of an electrical cycle or $\frac{1}{12}$ of a revolution. The instantaneous current values were obtained from a dynamic simulation of the drive system [4] from which the current waveform shown in Fig. 7 was obtained. It will be seen that under the assumed constant speed condition, the magnetic field produced by the magnets rotates at a constant speed, whilst the magnetic field produced by the stator windings rotates in incremental steps as commutation events occur and the phase currents vary. As a consequence, the resultant magnetic field on load is nonuniformly.

At time $t = 0$ s only phases B and C are conducting with phase A about to be commutated on, while at $t =$ 1.666 ms only phases A and C are conducting. During the intervening time interval the axis of the armature reaction field rotates an incremental angle of 30° elec. whilst the

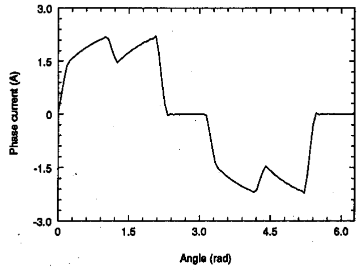

Fig. 7. Typical stator winding current waveform (1500 r/m).

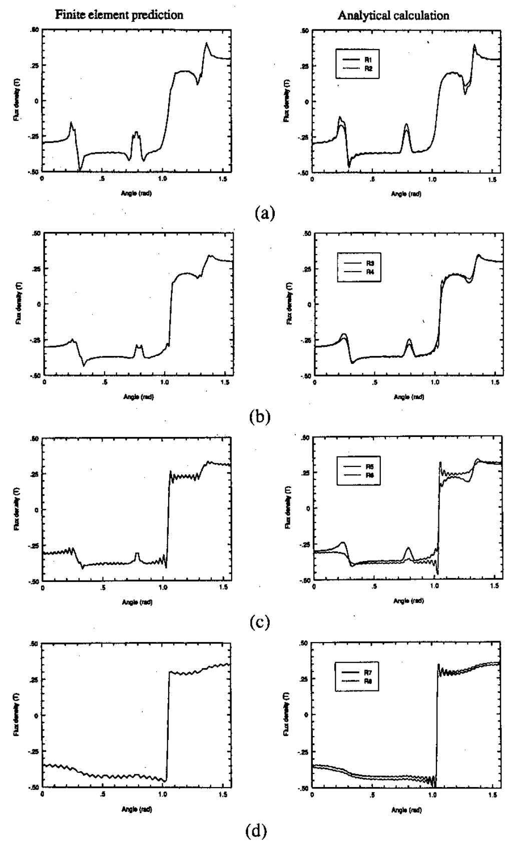

Fig. 8. Comparison of resultant field distribution at different radii. Time = 0s. (a) $r = R_s^-$. (b) $r = R_s - g^+$. (c) $r = R_s - g^-$. (d) $r = R_s - g - h_m^+$. The definition of radii is given in Table I of Part I.

rotor has rotated 15° mech. However, during the next time interval to $t = 3.333$ ms no further commutation event occurs. Therefore during the next 15° mech. rotation of the rotor the axis of the armature reaction field remains essentially fixed at the position it occupied at $t = 1.666$ ms. However the magnitude of the field changes as the instantaneous phase currents vary. Therefore under a load condition the angle between the axes of the open-circuit field and armature reaction field components effectively varies from 120° elec. at the beginning of a commutation period to 60° elec. at the end of the period.

In Figs. 8, 9, and 10 predictions from the proposed analytical method are compared with flux density distributions derived from the finite element predictions of Fig. 6. The open-circuit and armature reaction component field distributions as well as the resultant load field distributions all show excellent agreement, particularly bearing in mind comments made in part I–III [5]–[7] regarding the effects of finite element discretization, flux focusing and flux leakage, and the sources of oscillations in the predicted field distributions. However, it should also be noted that the particular motor which has been analyzed has a relatively low operating flux density, so that it is only lightly saturated.

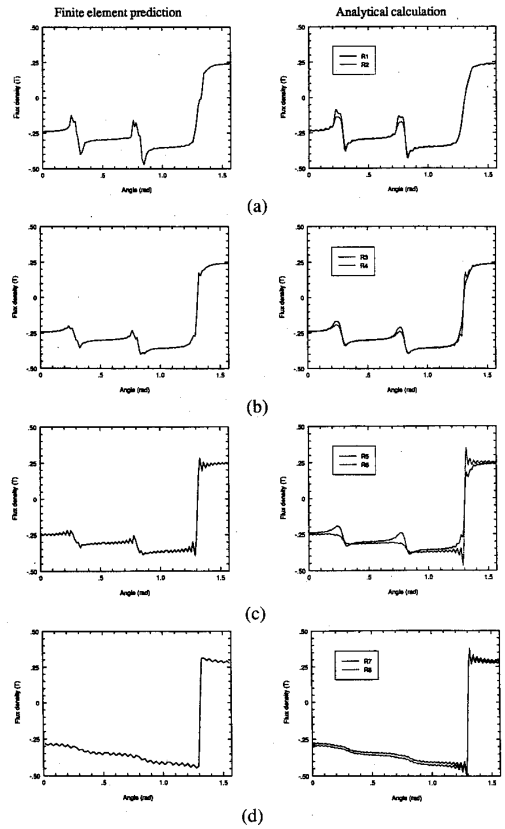

Fig. 9. Comparison of resultant field distribution at different radii. Time = 1.66 ms. (a) $r = R_s^-$. (b) $r = R_s - g^+$. (c) $r = R_s - g^-$. (d) $r = R_s - g - h_m^+$. The definition of radii is given in Table I of part I [5].

V. Conclusions

An analytical technique, which is suitable for both internal and external rotor topologies of radial-field brushless dc motors having a surface-mounted magnet rotor, has been developed for predicting the airgap field distribution under any specified load condition. It accounts implicitly for the stator winding current waveform and the effects of stator slot openings. The technique is based on a 2-dimensional analysis, in polar coordinates, of the airgap-permanent magnet region, and combines an analytical calculation of the stator current waveform and the armature reaction field with an analytical prediction of the open-circuit field produced by the magnets. Fundamental

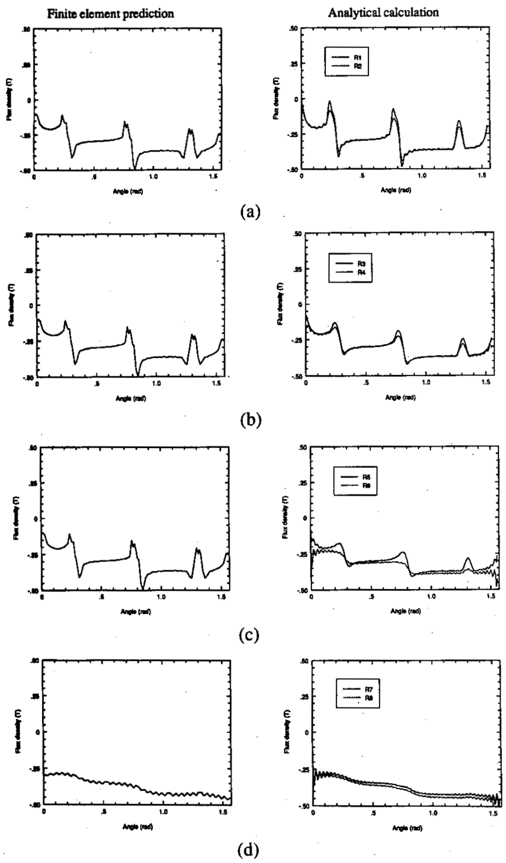

Fig. 10. Comparison of resultant field distribution at different radii. Time = 3.33 ms. (a) $r = R_s^-$. (b) $r = R_s - g^+$. (c) $r = R_s - g^-$. (d) $r = R_s - g - h_m^+$. The definition of radii is given in Table I of part I [5].

to the calculation of the instantaneous field distributions is a knowledge of the relative position of the open-circuit and armature reaction field components, the permeance function, and the back-emf and current waveforms. Predicted field distributions all show excellent agreement with results from corresponding finite element analyses, despite necessitating some degree of idealization, mainly with regard to the assumption of infinitely permeable iron. However, the technique is likely to be preferred during the initial stages of the machine/system design process, and will, therefore, complement more refined numerical techniques which may be used at a later stage.

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Appendix
EMF Induced in Windings of a 3-Phase Brushless dc Motor

The induced emf waveform can be calculated from a knowledge of the open-circuit magnetic field distribution and the stator winding distribution.

For a slotless motor topology the open-circuit airgap field is given by

$$B_{open\text{-}circuit}(\theta, r) = B_{magnet}(\theta, r) = \sum B_n \cos np\theta. \quad (28)$$

The flux density distribution at the stator bore, which determines the emf induced in the windings, is

$$B_{open\text{-}circuit}(\theta) = B_{magnet}(\theta, r)|_{r=R_s}$$

where

$$\theta = \alpha - \alpha_{ma} \quad (29)$$

and α and θ are the angular position referred to the stator and the rotor, respectively, α_{ma} being their relative position which depends on the speed of rotation, i.e.,

$$\alpha_{ma} = \omega_r t + \theta_o \quad (30)$$

where θ_o is an initial angle, which depends on the commutation angle.

When a motor is slotted an average value of relative permeance $\tilde{\Lambda}_o$ can be used to account for the reduction of the effective flux density.

Therefore the open-circuit flux density distribution at the stator bore of both slotted and slotless motors is given by

$$B_{open\text{-}circuit}(\alpha, t) = \tilde{\Lambda}_o \sum_n B_n \cos np(\alpha - \alpha_{ma}). \quad (31)$$

Hence the flux linking a stator coil is calculated from

$$\psi = \int_{-(\alpha_y/2)}^{\alpha_y/2} B_{open\text{-}circuit}(\alpha, t)\, R_s l_{ef}\, d\alpha \quad (32)$$

where α_y is the winding pitch angle, R_s is the radius of the stator bore, and l_{ef} is the effective axial length of a stator coil.

Hence

$$\begin{aligned}\psi &= \tilde{\Lambda}_o \sum_n B_n R_s l_{ef} \int_{-(\alpha_y/2)}^{\alpha_y/2} \cos np(\alpha - \alpha_{ma})\, d\alpha \\ &= \tilde{\Lambda}_o \sum_n 2B_n R_s l_{ef} \int_0^{\alpha_y/2} \cos np\alpha \cos np\alpha_{ma}\, d\alpha \\ &= \tilde{\Lambda}_o \sum_n \frac{2B_n R_s l_{ef}}{np} \sin np\frac{\alpha_y}{2} \cos np\alpha_{ma} \\ &= \tilde{\Lambda}_o \sum_n \frac{\Phi_n}{np} K_{dn} \cos np\alpha_{ma} \quad (33)\end{aligned}$$

where

$$\Phi_n = 2B_n R_s l_{ef} \tilde{\Lambda}_o \quad (34)$$

and

$$K_{dn} = \sin np\frac{\alpha_y}{2}.$$

Therefore the emf induced in each turn of a coil is

$$\begin{aligned}e &= -\frac{d\psi}{dt} = \tilde{\Lambda}_o \sum_n 2B_n R_s l_{ef} \omega_r K_{dn} \sin np\alpha_{ma} \\ &= \sum_n \omega_r \Phi_n K_{dn} \sin np\alpha_{ma} \quad (35)\end{aligned}$$

Hence for a distributed multi-pole winding, the induced emf per phase is obtained as

$$e = \sum_n \omega_r \Phi_n W K_{dpn} \sin np\alpha_{ma} = \sum_n E_n \sin np\alpha_{ma} \quad (36)$$

where

$$E_n = \omega_r \Phi_n W K_{dpn} \quad (37)$$

in which W is the number of turns per phase; $K_{dpn} = K_{dn} K_{pn}$ is the winding factor, K_{pn} is the winding distribution factor, and K_{dn} is the winding pitch factor.

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